



Koneru Lakshmaiah Education Foundation

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DATA SCIENCE(24AD3107)

2026-27, ODD Semester

ABSTRACT

Sec No-08	Team No-18	Project Title- “EduInsight” – Student Performance & Dropout Full-Cycle Analytics System
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2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Name
Data Ingestion, Preprocessing & Quality Assessment Module	2420030571	I. Adhithya
Exploratory Data Analysis & ETL Integration Module	2420030568	Y. Avinash
Machine Learning, Risk Prediction & Streamlit Dashboard Module	2420030364	B. Charan

1. Abstract:

EduInsight is an end-to-end student performance and academic risk analytics system designed to help educational institutions identify factors influencing student outcomes and detect students who may require academic support. The system transforms raw student data into meaningful insights through a complete data analytics and machine learning pipeline. The project uses student-related attributes such as study time, attendance, previous academic performance, failures, demographic information, and other relevant factors to analyze academic behavior and performance.

The proposed system begins with automated data ingestion and preprocessing, where missing values, duplicate records, inconsistent data, and outliers are identified and appropriately handled. A data quality assessment module evaluates the dataset based on completeness, consistency, validity, uniqueness, and other quality parameters. Exploratory Data Analysis (EDA) is then performed using statistical methods and visualizations to identify relationships between student characteristics and academic performance. The system can also integrate multiple datasets, such as student information, attendance records, and examination results, using an **ETL-based data integration process**.

Machine learning algorithms including **Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbors** are applied to predict student performance and classify students into different academic risk levels such as Low, Medium, and High Risk. Model performance is evaluated using appropriate metrics, and the most effective model is selected for deployment. The system also provides interpretable insights by identifying important factors contributing to a student's predicted performance or risk level.

An interactive Streamlit dashboard serves as the primary interface, allowing users to upload or explore datasets, monitor data quality, visualize academic trends, compare machine learning models, analyze individual students, and generate predictions. The dashboard provides actionable information that can assist educators and administrators in identifying struggling students at an early stage and planning appropriate interventions. Overall, EduInsight demonstrates how data engineering, exploratory analytics, machine learning, and interactive visualization can be integrated into a practical educational decision-support system.

Guide Name and Signature (remarks if any) : Dr. N. Sunanda